UNICC AI Safety Council

Frontend Interface and Deployment Layer for Council-of-Experts AI Safety Evaluation

Applied Project Final Report

By

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Declaration

I, Yuze Wu, declare that this project report submitted by me to the School of Professional Studies, New York University, in partial fulfillment of the requirement for the award of the degree of Master of Science in Management and Systems, is a record of project work carried out by me under the guidance of Dr. Andrés Fortino and Joseph X. Ng. I grant powers of discretion to the Division of Programs in Business, School of Professional Studies, and New York University to allow this report to be copied in part or in full without further reference to me. This permission covers only copies made for study purposes or for inclusion in Division of Programs in Business, School of Professional Studies, and New York University research publications, subject to normal conditions of acknowledgment. I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Acknowledgments

I would like to thank Dr. Andrés Fortino and Joseph X. Ng for their guidance, feedback, and steady emphasis on making the Spring 2026 UNICC AI Safety Lab evaluable as a real product rather than merely a classroom demonstration. Their support consistently pushed the work toward clearer scope, stronger documentation, and a more disciplined connection between governance goals and operational usability.

I am equally grateful to the United Nations International Computing Centre (UNICC) and to the broader project stakeholders who defined the business need for a trustworthy AI safety evaluation workflow inside the AI Sandbox. Their problem framing made it clear that technical sophistication alone is not sufficient; evaluators also need a usable, auditable, and stakeholder-ready interface.

I also want to acknowledge my teammates, Jun Kim and Wesley Koe, whose work on the broader council architecture, module integration, and system logic made the final team product possible. My own report focuses on interface, testing, evaluator-readiness, and deployment, but these contributions only became meaningful because they were integrated into a coherent multi-module product.

Finally, I would like to thank the faculty and instructors in the Management and Analytics program at NYU School of Professional Studies. Courses in analytics, systems design, project management, and communication directly shaped the way I approached this capstone as both an analytical exercise and a deployable product problem.

Abstract

International organizations are under growing pressure to use artificial intelligence in operational environments, yet they often lack a consistent and auditable way to determine whether an AI system is safe enough to deploy. This project addresses that gap for the United Nations International Computing Centre (UNICC) by developing the interface, testing, and deployment layer of the Spring 2026 UNICC AI Safety Lab, a council-of-experts evaluation system designed to review AI agents before production use inside the UNICC AI Sandbox. This work specifically corresponds to Project C, focusing on UX/UI design, evaluator workflow, and deployment readiness within the broader multi-module system.

The proposed solution operationalizes a multi-module safety architecture through an evaluator-facing workflow that accepts dynamic product inputs, surfaces three distinct expert assessments, presents deliberation and council synthesis clearly, and generates a final stakeholder-readable recommendation. The proof of concept specifically aimed to make the broader team solution usable for first-time reviewers, robust under limited infrastructure, and aligned with the course evaluation standard built around the VeriMedia benchmark case.

Methods included translating user and evaluator requirements into interface flows, integrating governance, threat, and behavioral expert outputs into a unified presentation layer, designing fallback execution for no-key or low-resource conditions, and validating the workflow with VeriMedia-specific analysis as well as structured council scenarios. The resulting system was able to accept dynamic inputs, preserve module independence, support explicit synthesis into APPROVE/REVIEW/REJECT verdicts, and present outputs in a way that non-technical stakeholders could follow without manually reading raw JSON.

Key results include a deployable browser-based prototype, a council visualization workflow that highlights structured disagreement rather than hiding it, and a demonstration path that remained functional even when heavy model loading or live credentials were unavailable. These results support the conclusion that evaluator-readiness is not a cosmetic layer placed on top of AI governance; it is a core part of whether governance can be practiced consistently at all.

The project implies that organizations such as UNICC can improve the trustworthiness of their AI review process by pairing governance frameworks with an accessible testing workflow and explicit decision logic. Future work should extend the prototype to live DGX deployment, persistent audit logging, broader multilingual and multimodal testing, and deeper human-in-the-loop review controls. The main contribution of this project is a practical, professionally documented blueprint for making a council-of-experts AI safety system usable, testable, and presentation-ready in a mission-critical institutional setting.

Abbreviations and Definitions

Abbreviation	Full Term / Definition
AI	Artificial Intelligence
API	Application Programming Interface
DGX	NVIDIA DGX Spark infrastructure used by NYU for controlled AI experimentation
GRC	Governance, Risk, and Compliance
LLM	Large Language Model
MoE	Mixture of Experts / multi-module inference ensemble
PoC	Proof of Concept
SLM	Small Language Model
TOC	Table of Contents
UNICC	United Nations International Computing Centre
UX	User Experience
VeriMedia	The standard benchmark AI agent used by the Spring 2026 evaluator for repository testing

Introduction

Large language models (LLMs) and AI agents are rapidly moving from experimental tools into operational systems that support workflow automation, content generation, decision-making, and coordination across organizations. While these technologies offer clear efficiency gains, they also introduce new challenges around reliability, safety, usability, and governance. Organizations can define high-level responsible AI principles, but they often struggle to translate those principles into a structured and repeatable evaluation process that evaluators can actually execute.

This project aims to demonstrate how these challenges can be addressed through a proof of concept developed within the Spring 2026 UNICC AI Safety Lab. While the broader team product integrates multiple expert modules into a council-of-experts safety evaluation system, this report specifically focuses on Project C: UX/UI design, testing workflow, and deployment readiness. The emphasis is on making AI safety evaluation practical, operational, and usable in real institutional settings

Problem

The specific problem addressed in this project is the lack of a usable and evaluator-ready interface layer for AI safety evaluation within the UNICC AI Sandbox. A council-of-experts architecture can generate multiple safety perspectives, but it becomes ineffective if evaluators cannot reliably submit inputs, understand expert outputs, follow the synthesis logic, or interpret the final decision. Without a structured interface workflow, AI governance remains abstract, inconsistent across reviewers, and difficult to apply in practice.

Approach

To address this problem, this project develops an evaluator-facing testing, interface, and deployment layer built around the council-of-experts architecture. The system enables users to submit AI systems, such as through a GitHub repository or structured input, view independent expert evaluations, and understand how these perspectives are synthesized into a final council decision. The goal is to operationalize AI safety evaluation into a clear, repeatable, and presentation-ready workflow.

Core Technology

The core technology is a browser-based AI safety evaluation interface integrated with a council-of-experts system. At the backend level, the system relies on structured JSON outputs, predefined arbitration rules, and expert-specific evaluation schemas to maintain independence between governance, threat, and behavioral assessments while producing a unified recommendation.

At the interface level, the system translates complex evaluation logic into a user-friendly workflow. It uses step-based progression, expert visualization cards, structured summaries, and council decision displays to help non-technical users interpret results. In addition, the system incorporates a graceful degradation mechanism, allowing evaluations and demonstrations to proceed even when full model infrastructure, live credentials, or high-resource inference conditions are unavailable. This ensures that testing and demonstration remain possible under real-world constraints.

Benefits

The proposed approach provides several practical and institutional benefits. First, it reduces friction for evaluators through a structured interface workflow that simplifies both input submission and result interpretation, allowing first-time users to engage with the system without requiring deep technical knowledge. Second, it increases transparency by explicitly surfacing differences across expert modules, enabling users to understand not only the final decision but also the reasoning and disagreements behind it. In addition, the system improves consistency and reproducibility through a standardized evaluation workflow, ensuring that similar inputs are assessed in a comparable manner across different reviewers and use cases. By making decision logic visible and structured, the interface also enhances trust in the evaluation process, which is particularly important in governance contexts. Finally, the system supports deployment readiness by functioning effectively under both full and constrained environments, allowing evaluation workflows to remain stable even when infrastructure limitations are present.

Research Question

The research question explored in this proof of concept is: how can a UX/UI, testing, and deployment workflow operationalize a council-of-experts AI safety system into a usable, evaluator-ready, and institutionally credible evaluation process within the UNICC AI Sandbox?

To address this question, the report first defines the project objectives and evaluation metrics, then examines alternative solution approaches, reviews relevant literature, and presents the system design and methodology. It subsequently reports the results of the proof of concept before concluding with key insights, limitations, and directions for future work.

Contribution

The primary contribution of this project is the development of an operational interface layer that transforms a multi-expert AI safety architecture into a usable and evaluable product. This contribution is reflected in the design of a structured submission workflow that standardizes how AI systems are introduced into the evaluation process, the clear visualization of independent expert outputs that preserves their distinct perspectives, and the transparent presentation of deliberation and arbitration mechanisms that lead to the final decision. In addition, the project demonstrates a deployment-ready evaluation workflow that remains functional under real-world constraints, providing a practical blueprint for integrating AI safety governance into institutional settings.

Sponsor

The project sponsor is Dr. Andrés Fortino, Clinical Associate Professor at the NYU School of Professional Studies. The project is conducted in collaboration with the United Nations International Computing Centre (UNICC), a global organization providing digital solutions, cybersecurity, and AI services to UN agencies. UNICC operates within the international public-sector technology services industry, supporting over 90 organizations across the United Nations system.

Importance of Project

This project is important to the sponsor because UNICC requires a reliable and defensible method to evaluate AI systems before deployment. Without a structured evaluation workflow, AI adoption may lead to inconsistent reviews, weak documentation, and increased operational risk.

By developing a usable AI safety evaluation interface, this project helps UNICC move from experimental AI usage toward standardized, auditable, and governance-aligned deployment practices, supporting responsible AI adoption across the UN system.

Project Objectives and Metrics

Goal of the Project

The goal of this project was to design, test, and document the interface, evaluation workflow, and deployment pathway for the Spring 2026 UNICC AI Safety Lab so that a reviewer could submit a real AI system, inspect three distinct expert perspectives, and receive a clear council recommendation suitable for governance use, product demonstration, and formal repository evaluation.

Project Objectives

Objective 1 — Operationalize evaluator-ready submission workflow

The system must support dynamic input submission of AI systems, such as through GitHub repositories or structured descriptions, and automatically normalize inputs into the council evaluation pipeline without requiring manual preprocessing. From an interface perspective, this workflow must be accessible to evaluators without requiring direct interaction with backend code or raw API calls.

Objective 2 — Preserve and display multi-agent evaluation outputs

The system must maintain independent outputs from governance, threat, and behavioral expert modules, each producing structured evaluation results. The interface must clearly present these outputs with distinct visual separation, including risk levels, recommendations, and rationale, ensuring that evaluators can interpret differences across expert perspectives.

Objective 3 — Implement structured council decision logic and readable synthesis

The system must apply explicit arbitration logic to aggregate expert outputs into a final council decision, including risk-tier classification and justification. The interface must translate this synthesis into a clear and structured presentation, allowing non-technical users to understand both the final verdict and the reasoning process behind it.

Objective 4 — Deliver a deployable, auditable, and human-usable evaluation interface

The final system must provide a web-based interface that supports testing, demonstration, and governance review. It must include structured outputs such as risk-tier classification, visualized evaluation results, and a clear audit trail aligned with UNICC requirements. The system must remain functional under both full and constrained execution environments, ensuring deployment readiness within the NYU AI Sandbox.

Project Metrics

Project success was evaluated using both academic and product-oriented criteria. At the academic level, success required a complete and professionally structured report aligned with the capstone template. At the product level, success was defined by whether the system could accept dynamic inputs, produce structured multi-expert outputs, generate a final council decision, and present results in a way that is interpretable to first-time evaluators.

Operational metrics included successful input acceptance, clarity of module distinction, completeness of council synthesis, readability of outputs, and system stability under constrained execution conditions. These metrics reflect the core requirement that the system must not only function technically, but also support a usable and auditable AI safety evaluation workflow.

Alternate Solutions Evaluated

Several solution paths were considered before the final implementation direction was selected. Because my role focused on interface, testing, and deployment rather than on a single expert model, the alternatives were evaluated primarily in terms of evaluator usability, visibility of decision logic, and feasibility within the Spring 2026 capstone structure.

The first alternative was a backend-only review workflow. Under this design, evaluators would submit an AI system through a command line or raw API call and receive a JSON payload containing the expert outputs and final council decision. This option appeared attractive at first because it minimized UI engineering effort and kept the system close to the underlying code. However, it quickly became clear that a backend-only path would fail the spirit of the project. It would force non-technical reviewers to interpret nested JSON structures, would hide the difference between raw module execution and readable governance reporting, and would make live demonstrations fragile.

The second alternative was a single-judge interface. In this design, the user experience would be optimized around one consolidated safety reviewer rather than three independent experts and a council synthesis layer. From a product simplification perspective, this was tempting because a single verdict is easier to present than structured disagreement. However, this approach conflicted with the Spring 2026 project definition. The entire purpose of the semester's architecture was to preserve the distinct logic of the inherited Fall 2025 solutions and operationalize them as a council of experts. A single-judge interface would therefore have reduced complexity at the cost of undermining the core research and product hypothesis.

The third alternative was a static demonstration interface, essentially a polished walkthrough designed around a fixed scenario with pre-generated outputs. Such a design would have been visually impressive and relatively easy to stabilize for presentation day. However, the formal evaluator guide explicitly penalized systems that relied on hardcoded demos rather than dynamic input. A static interface would also weaken the credibility of the project by making the product look more like a slideshow than a working safety review workflow.

The fourth alternative was a full enterprise-style dashboard with persistent databases, authentication layers, historical comparison views, and advanced administrative controls. This option aligned with long-term institutional needs, but it exceeded the realistic scope of a semester-long proof of concept. Pursuing a heavy enterprise stack would have shifted the project away from the core deliverable—making the multi-expert workflow evaluable and readable—and toward infrastructure work that would be difficult to complete credibly within the capstone timeline.

The final solution therefore positioned itself between oversimplification and overengineering. It aimed to be dynamic enough to accept real benchmark inputs, explicit enough to preserve expert independence and council synthesis, and lightweight enough to deploy, test, and demonstrate under real capstone constraints.

Solution Evaluation Criteria

Four criteria were used to compare the alternatives.

First, the solution needed evaluator fitness. Because the project would be judged by a first-time evaluator using VeriMedia as a standard input, the system had to support a review experience that did not require code edits, hidden setup steps, or manual schema conversion.

Second, the solution needed architectural faithfulness. It had to preserve the actual value proposition of the Spring 2026 AI Safety Lab: three expert modules, explicit critique or synthesis, and a final council verdict. Any design that collapsed that structure into a single opaque answer failed the project's conceptual brief.

Third, the solution needed communication quality. It was not enough for the system to compute a result. The result had to be readable by a non-technical stakeholder, connected to the underlying expert outputs, and suitable for governance discussion.

Fourth, the solution needed delivery feasibility. The capstone timeline, infrastructure limits, and demonstration requirements meant that the chosen design had to be implementable within one semester and robust enough to survive missing keys, limited resources, or presentation-day variability.

Selection Rationale

The final selection was a lightweight but dynamic web-based evaluation workflow built specifically around the council-of-experts architecture. This choice was driven by the fact that it was the only option that scored well across all four criteria.

Compared with a backend-only design, the selected solution dramatically improved evaluator fitness. It made the workflow legible without forcing a stakeholder to read raw JSON or inspect source code. This mattered both for the formal scoring rubric and for the actual institutional use case. A governance tool that only developers can interpret does not meaningfully solve the organization's review problem.

Compared with a single-judge interface, the chosen design remained faithful to the Spring 2026 architecture. It displayed governance, threat, and behavioral assessments separately and then showed how the council reached a final recommendation. This structure preserved the project's central hypothesis that diversity of perspective leads to more robust AI safety evaluation. More importantly, it turned that architectural principle into something visible. Users could see where the experts aligned, where they diverged, and why human review remained necessary.

Compared with a static demo, the selected approach maintained product integrity. It still supported presentation-ready walkthroughs, but it did so without giving up dynamic input acceptance. This was especially important given the explicit warning in the evaluator guide that a hardcoded demo would score poorly. In practical terms, the system was designed so that a benchmark repository such as VeriMedia could be entered as a live input while the product still retained a stable and polished presentation layer.

Compared with a full enterprise dashboard, the chosen solution was more realistic and more strategically disciplined. It prioritized the features that directly advanced the proof of concept: input handling, module display, synthesis readability, fallback execution, and deployment accessibility. Features such as persistent databases, multi-tenant administration, and deep authentication controls were deferred because they did not contribute enough to the immediate semester objective relative to their engineering cost.

The selection also reflected a broader product judgment. In institutional AI governance, the highest-value feature is often not the most complex technical capability; it is the ability to make the system's reasoning visible and actionable. The chosen solution was therefore selected because it maximized the ratio of governance value to implementation complexity. It did not pretend to be the final enterprise platform UNICC might eventually operate. Instead, it served as a disciplined proof of concept that demonstrated how the broader AI Safety Lab could be reviewed, presented, and iterated in a controlled environment.

Literature Survey

Introduction

The literature relevant to this project sits at the intersection of AI governance, model evaluation, explainability, and human-centered system design. The goal of this survey is not merely to summarize ethical principles. It is to identify what prior work says about turning abstract governance requirements into repeatable institutional workflows. That concern is especially important for a project such as the UNICC AI Safety Lab, where the problem is not only whether unsafe outputs exist, but whether an organization can review AI products consistently before release.

This review therefore proceeds in six parts. It first examines the broader industry context for AI governance in mission-critical institutions. It then reviews the problem of implementation gaps between principles and practice. Next, it surveys the types of proposed solutions that attempt to operationalize AI review. It then summarizes the most relevant technologies, especially LLM-as-judge systems, multi-expert evaluation, explainability, and adversarial testing. The review closes with concrete use cases and a conclusion that links the literature directly to the design choices made in this project.

The Industry

The adoption of generative AI inside organizational workflows has moved quickly from experimentation to operational deployment. Foundation models are now used for summarization, document analysis, classification, conversational support, and decision assistance in both private and public institutions (Bommasani et al., 2021). As these systems become embedded in real work, the industry conversation has shifted from raw capability to questions of trust, assurance, and governance. Brundage et al. (2020) emphasize that organizations need verifiable claims about model behavior rather than vague assurances about safety. This shift is particularly visible in highly regulated or mission-critical environments, where failure has consequences beyond product inconvenience.

The United Nations context intensifies these pressures. AI deployed in humanitarian, operational, and policy settings can affect vulnerable populations, resource allocation, and organizational legitimacy. The UNICC memorandum for Spring 2026 framed the challenge as one of pre-deployment testing inside the AI Sandbox: systems should be reviewed before production integration, through auditable and transparent mechanisms, without reliance on black-box external APIs. That framing aligns with the broader governance trend captured by the NIST AI Risk Management Framework and ISO/IEC 42001, both of which place strong emphasis on lifecycle governance, documentation, risk treatment, and organizational accountability (NIST, 2023; ISO/IEC, 2023).

The industry therefore increasingly recognizes that AI governance is not only a policy topic. It is also a systems problem. Institutions need workflows, artifacts, interfaces, and evidence structures that translate governance expectations into review practice. That broader realization is the context in which the UNICC AI Safety Lab should be understood.

The Problem

A recurring finding in the literature is that organizations publish AI principles more easily than they operationalize them. Jobin et al. (2019) showed that fairness, transparency, accountability, and privacy recur across many AI ethics frameworks, yet the existence of shared principles does not automatically produce consistent implementation. Mittelstadt (2019) argues that principles often remain too abstract to guide action unless they are connected to enforceable procedures, tools, and accountability structures. Morley et al. (2020) reach a similar conclusion in their survey of AI ethics tools: organizations often face a translation problem, moving from "what we value" to "how we will test and document it."

This implementation gap becomes even more serious with large language models. Ji et al. (2023) synthesize the literature on hallucination and highlight the difficulty of validating stochastic systems through conventional software testing alone. Weidinger et al. (2022) propose a taxonomy of risks posed by language models, emphasizing that harm can arise through toxicity, bias, misinformation, misuse enablement, and downstream operational effects. These are not failure modes that can be reduced to one simple pass/fail check. They require layered evaluation and often involve judgment under uncertainty.

A second problem identified in the literature concerns interpretability and user trust. Doshi-Velez and Kim (2017) argue that interpretability is not just a technical add-on; it is central to whether human decision-makers can use model outputs appropriately. Ribeiro et al. (2016) make a similar point in more practical terms: users need explanations they can inspect if they are expected to trust and act on complex systems. For an institutional AI safety product, that means the review workflow must communicate more than an opaque score. It must expose reasoning pathways, uncertainty, and triggers for human escalation.

A third problem is adversarial robustness. Ganguli et al. (2022) and Perez et al. (2022) show that red-teaming and adversarial probing are necessary because many unsafe behaviors only appear when a system is stressed intentionally. If organizations rely only on ordinary-use testing, they are likely to miss the kinds of harmful behavior that matter most in sensitive contexts. The implication is that a credible AI safety workflow needs both structured evaluation logic and interface-level support for seeing what was tested and why.

The Proposed Solution

The literature increasingly supports hybrid approaches that combine governance frameworks, structured evaluation logic, and human-readable reporting. The EU AI Act provides a strong legal model for risk-based classification and control intensity, especially for higher-risk uses (European Union, 2024). NIST (2023) provides a process model built around mapping, measuring, managing, and governing risk. ISO/IEC 42001 provides an organizational management-system structure that anchors responsibility, continuous improvement, and documented control processes (ISO/IEC, 2023). No single framework fully solves the implementation problem by itself, but together they suggest that the most effective institutional solution is a layered one.

The proposed solution in this project follows that layered logic. At the system level, the broader Spring 2026 product adopted a council-of-experts model so that safety review would not depend on a single monolithic judge. At the workflow level, my contribution translated that logic into a submission-and-review product that an evaluator could actually use. This design is consistent with the literature's emphasis on auditable AI review. Binns (2018) argues that accountable algorithmic systems require processes that support public reason rather than hidden calculation. Similarly, Brundage et al. (2020) emphasize the need for supporting evidence and interpretable claims when advanced AI systems are evaluated.

What is distinctive about the proposed solution is that it treats usability as part of governance. Instead of assuming that governance happens elsewhere and the interface is a downstream presentation layer, the project integrates submission, expert display, synthesis, and escalation into one workflow. This is supported by the literature because the implementation gap is not only technical; it is organizational. A control is not really a control if users cannot execute it consistently.

The Technology

Several technology streams informed the project. First is the rise of LLM-as-judge methods. Zheng et al. (2023) show that strong models can be used to assess model outputs at scale, but they also caution that judge systems can introduce their own biases and must be structured carefully. This insight matters because the Spring 2026 AI Safety Lab built on prior judge-based systems rather than abandoning them. The council approach can be understood partly as a response to the limitations of single-judge evaluation: distinct perspectives can reduce the risk that one reasoning pattern becomes the sole authority.

Second is the literature on adversarial testing and red teaming. Ganguli et al. (2022) and Perez et al. (2022) show that harmful behaviors often require deliberate probing to surface. These findings justify the project’s emphasis on testing readiness, scenario input design, and explicit review of disagreement. If the system only shows a final verdict, users lose access to the very evidence that makes adversarial evaluation useful.

Third is explainability and model documentation. Doshi-Velez and Kim (2017) provide the conceptual case for interpretability, while Mitchell et al. (2019) and Gebru et al. (2021) show

how model cards and datasheets can turn documentation into a governance artifact. The UNICC workflow is not identical to these tools, but it shares their logic: structured metadata and readable summaries are part of what makes an evaluation useful.

Finally, the technology literature suggests that orchestration matters as much as raw model performance in institutional settings. A council architecture, especially one with explicit arbitration, creates a way to expose different risk signals without pretending that all safety concerns are reducible to a single scalar metric. That orchestration logic strongly influenced the interface design choices in this project.

Use Cases

The most important use case for the Spring 2026 project was the evaluation of VeriMedia, the benchmark system named in the evaluator guide. VeriMedia is a Flask-based AI agent that analyzes text, audio, and video content for toxicity using GPT-4o and Whisper. The guide specifically instructed teams to reference VeriMedia's Flask architecture, file-upload surface, GPT-4o backend, and lack of an authentication layer in their evaluations. This is an ideal example of why a readable workflow matters. A reviewer must connect technical observations about architecture and risk surface to a final institutional recommendation.

A second use case comes from humanitarian or operational AI assistants that may produce sensitive summaries or recommendations. In such settings, structured disagreement between experts may be more informative than superficial agreement. A threat-oriented expert may object to a capability that a governance-oriented expert views as acceptable under policy, while a behavioral expert may focus on overreliance or user misuse. A council interface allows these distinctions to remain visible.

A third use case is internal organizational review itself. Even when an AI system is not deployed to end users directly, institutions still need a repeatable process for triage, escalation, and sign-off. The product designed in this project supports that use case by creating a pathway from dynamic input to readable output without assuming that every reviewer is also a developer.

Conclusion

The literature converges on a clear conclusion: AI governance fails in practice when institutions stop at principles and do not build repeatable, inspectable workflows. Governance frameworks define what matters, but organizations still need evaluation systems that surface risk, support human review, and communicate outcomes in a credible form. The most relevant prior work therefore supports the core direction of this project: a council-of-experts architecture presented through a usable submission, review, and decision workflow.

For the UNICC AI Safety Lab specifically, the literature justifies three design choices. First, risk should be evaluated through multiple lenses rather than a single opaque judgment. Second, usability and readability are not cosmetic; they are necessary for institutional accountability. Third, fallback execution and careful reporting matter because real evaluation environments are constrained. These insights formed the conceptual foundation for the methodology and results presented in the next sections.

Approach and Methodology

Problem Statement and Research Question

The practical problem addressed by this project was not the design of a single expert model but the operationalization of the council review workflow. UNICC needed a mechanism that would allow a reviewer to submit an AI product, inspect three independent expert assessments, follow the synthesis logic, and receive a final recommendation that could support governance action. Without that mechanism, even a strong backend would remain difficult to validate, difficult to demonstrate, and difficult to use consistently.

The research question guiding the methodology was therefore: How can the interface, testing, and deployment layer of a multi-expert AI safety lab transform governance logic into an evaluator-ready workflow that supports real benchmark submission, readable deliberation, and defensible final decisions inside the UNICC AI Sandbox?

Proof of Concept Approach

The proof-of-concept approach treated the interface as a control surface for the broader council architecture. Rather than building an isolated front end disconnected from the system logic, the design organized the workflow around five steps: input submission, system profiling, expert review, deliberation display, and council decision. This made the product closer to an institutional review workflow than to a generic demo site.

A central design objective was evaluator readiness. The system had to support the benchmark review scenario defined in the evaluator guide, where a first-time reviewer would submit

VeriMedia and expect meaningful module outputs without editing configuration files. This requirement led to two important design choices. First, the interface supported dynamic input through a GitHub URL or structured product description rather than relying only on hardcoded scenarios. Second, the interface treated readability as part of the output schema. Module status, risk level, recommendation, and rationale all needed visible labels and stable presentation.

A second design objective was graceful degradation. The project environment could not assume persistent access to live production endpoints or large-scale inference resources. The proof of concept therefore preserved architecture even when heavy model loading was constrained. In the preserved council output schema, each expert still returned its own assessment fields, and the arbitration layer still produced a final council decision with human-review requirements. This allowed the testing and presentation workflow to remain valid even under sandbox portability mode.

Technology Trial Plan

The technology trial plan was designed around the actual Spring 2026 evaluation conditions rather than around an abstract laboratory benchmark. The primary validation case was VeriMedia because it was the standard repository that evaluators were instructed to use. VeriMedia was treated as a realistic external AI product with identifiable architecture and risk surface rather than as a synthetic classroom input.

The trial plan combined three forms of validation. The first was functional workflow validation. This asked whether the full submission-to-verdict path executed without friction. The second was output structure validation. This asked whether governance, threat, and behavioral expert outputs remained distinct and whether the final council recommendation could be read and justified. The third was presentation and deployment validation. This asked whether the system could still function credibly when infrastructure limitations prevented full heavy-model execution.

For functional testing, the interface was exercised in both dynamic-input and structured-scenario modes. Dynamic input mode focused on repository-based review, especially the submission of VeriMedia by URL. Structured-scenario mode focused on council outputs for safety-sensitive use cases where disagreement between experts was expected. The combination of both modes allowed the product to be tested for evaluator fitness while also validating the communication of disagreement, escalation, and human-review requirements.

For deployment validation, the interface was assessed under a no-key or constrained execution setting. The intent was not to hide limitations but to confirm that the council architecture could remain inspectable even when heavy inference was unavailable. This was particularly important because the evaluation guide explicitly warned teams not to rely on a live API key for the synthesis step.

Population/Data

The project did not depend on a traditional static dataset. Instead, it used two categories of evaluation inputs.

The first category consisted of repository-level product inputs, with VeriMedia serving as the primary benchmark case. In this setting, the "population" was AI products submitted for pre-deployment review. Relevant variables included architectural form, presence of file-upload surfaces, external model dependencies, authentication controls, and exposure points that could affect governance or threat posture.

The second category consisted of structured evaluation scenarios used to test the council presentation workflow, especially cases designed to produce expert disagreement. These scenarios were valuable because they made it possible to validate the deliberation and arbitration layers even when a full live model execution path was not available.

Independent variables in the trial included: input type (GitHub URL versus structured product description), execution mode (live or constrained portability mode), and scenario risk profile (lower-risk informational review versus higher-risk misuse or governance concern). Dependent variables included successful submission acceptance, completeness of module outputs, clarity of module distinction, readability of the final verdict, presence of explicit human-review triggers, and stability of the presentation flow.

Procedures

The procedures followed in the proof of concept were as follows.

First, product requirements were translated into a concrete front-end workflow. This included defining the submission states, expert card structure, council summary layout, and fallback behavior when live resources were constrained.

Second, the benchmark input requirements were incorporated into the interface design. Because VeriMedia had to be accepted dynamically, the submission form was designed around repository URL and structured-description pathways rather than around fixed examples only.

Third, the system profile layer was implemented conceptually so that reviewers could move from raw submission to safety-relevant observations. For VeriMedia, this meant identifying and communicating the implications of its Flask architecture, GPT-4o and Whisper dependency surface, file upload interface, and lack of authentication controls.

Fourth, the expert display workflow was aligned to the council schema. Governance, threat, and behavioral results were presented as separate modules with their own labels, statuses, and rationales.

Fifth, the deliberation and arbitration path was implemented so that module outputs could feed into a final APPROVE, REVIEW, or REJECT recommendation rather than remaining as disconnected judgments.

Sixth, the end-to-end workflow was tested in presentation-ready conditions, including constrained execution environments, to confirm that the product remained usable and readable under realistic capstone limitations.

Organizational Change Plan

The implementation of the proposed system represents not only a technical enhancement but also a shift in how AI safety evaluation is conducted within an institutional setting. The existing evaluation process relies heavily on manual review, individual interpretation, and unstructured workflows, which creates inconsistencies in decision-making and limits scalability. The introduction of a structured interface and council-based evaluation workflow requires a transition toward a more standardized, transparent, and reproducible approach to governance.

The evaluation process in this project was therefore designed to prioritize operational quality rather than focusing on a single model performance metric. The effectiveness of the system was assessed through its ability to support real evaluation workflows, including the successful acceptance of external AI inputs, the clear distinction between expert perspectives, and the production of a coherent and interpretable final decision. Particular attention was given to whether evaluators could understand and trust the evaluation output without requiring technical expertise, as this directly reflects the system's usability in an organizational context.

Data collection was conducted through structured walkthroughs of the product, inspection of benchmark scenarios, and systematic review of generated evaluation artifacts. For repository-based inputs, the analysis relied on the system's ability to interpret and process submitted information into a standardized evaluation schema. For structured scenarios, the focus was placed on the expert outputs and the final council decision, ensuring that the workflow produced consistent and interpretable results. Rather than relying on large-scale numerical benchmarking, the project emphasized artifact-based evidence, including interface outputs, decision summaries, and evaluation traces, as these elements directly reflect how the system would be used in practice.

The analysis combined qualitative assessment with operational validation. From a qualitative perspective, the evaluation examined whether the interface effectively communicated expert reasoning, highlighted potential risks, and preserved meaningful differences between expert opinions. From an operational perspective, the system was assessed based on its ability to execute a complete evaluation workflow, including dynamic input processing, independent expert assessment, and final decision synthesis. A key design principle of the system is that disagreement between expert modules should be preserved and made visible rather than prematurely resolved. In a governance context, such disagreement often reflects the complexity of real-world risk assessment, and making these differences explicit supports more informed and accountable decision-making.

From an organizational standpoint, the adoption of this system would require evaluators to shift from informal, experience-based review processes to a structured and evidence-driven workflow. This change introduces greater consistency and transparency but also requires users to adapt to standardized evaluation steps and outputs. To support this transition, the system is designed to minimize friction by providing intuitive input mechanisms, clear visual outputs, and interpretable decision summaries. As a result, the organizational change is not only technological but also procedural, aligning evaluation practices with the broader goal of operationalizing AI governance within the UNICC environment.

Results

Data Processing

The project's data processing layer was intentionally lightweight because the objective was not to train a new model from scratch. Instead, the system normalized incoming product descriptions and expert outputs into a structured review format. For repository-based inputs, this meant translating a product surface into evaluation-relevant attributes. For scenario-based inputs, it meant preserving expert-specific fields—status, risk, recommendation, rationale, and references—so that the council display remained stable.

In practice, the most important processing task was schema discipline. An interface that receives inconsistent output structures becomes unreliable quickly, especially when multiple expert modules are involved. The workflow therefore emphasized stable response categories and conservative handling of constrained-execution cases so that the final review artifact remained readable even when the backend environment was imperfect.

Findings

The first major finding is that dynamic input acceptance materially improved the credibility of the safety review workflow. Designing the submission path around GitHub URL or structured system description aligned the product with the actual evaluator workflow rather than with an internal-only demo flow. In practical terms, this meant the system could engage with VeriMedia as a real input rather than as a pre-baked example.

The second major finding is that module independence became much more legible once it was treated as a UI and reporting problem rather than only as a modeling problem. Governance, threat, and behavioral assessments were easier to compare when each module had its own presentation space, labels, and rationale summary. This mattered because one of the central risks in multi-agent systems is that apparent diversity can become cosmetic if all outputs are rendered in the same undifferentiated style.

The third major finding is that council synthesis was strongest when the final recommendation explicitly preserved disagreement. In the sample council outputs used during testing, the system did not force false consensus. Instead, it showed that a Threat Expert could recommend rejection, a Governance Expert could recommend escalation, and a Behavioral Expert could recommend revision—after which the council could issue a final REJECT verdict while still documenting why human review was required.

The fourth major finding is that graceful degradation improved deployment reliability. When heavy model loading or live credentials were unavailable, the product could still demonstrate the correct submission logic, expert categories, council schema, and final decision behavior. This improved the product's robustness for presentations, walkthroughs, and evaluation under constrained conditions.

Summary Statistics

At the overall level, the prototype successfully achieved the primary operational objectives of the project and demonstrated clear proof-of-concept viability. The system was able to accept benchmark-style inputs, preserve distinct expert-level outputs, and synthesize these into a final council decision within a structured workflow. Importantly, the evaluation process remained functional even under constrained execution conditions, indicating robustness at the interface and deployment layers. These outcomes collectively validate that the proposed council-based architecture can operationalize AI safety evaluation into a usable and institutionally credible workflow, consistent with the UNICC objective of developing an auditable, multi-module safety testing system .

From a quantitative perspective, the project did not rely on large-sample statistical benchmarking, but instead on the consistency and completeness of structured evaluation outputs. Each council run reliably generated three independent expert assessments alongside a final synthesized verdict, demonstrating stable execution of the multi-agent pipeline. In representative structured-disagreement scenarios, the Governance Expert identified policy and neutrality concerns (Caution / High), the Threat Expert flagged critical misuse risks (Fail / Critical), and the Behavioral Expert highlighted user-level risks (Caution / Moderate). These differentiated outputs were consistently aggregated into a final council decision of Reject with Critical risk and a mandatory human-review requirement. While not framed as formal statistical testing, this repeatability functions as process-level validation, showing that the system consistently preserves expert disagreement and translates it into a deterministic institutional recommendation. In addition, baseline evidence inherited from Fall 2025 testing workflows supports the broader feasibility of automation, indicating that structured evaluation pipelines can reduce review time from hours to seconds, reinforcing scalability and operational efficiency.

Qualitative observations further support these findings by highlighting how users interacted with and interpreted the system. Presenting the evaluation as a staged workflow—submission, expert review, deliberation, and council decision—significantly improved user comprehension and reinforced the perception that the final output was the result of a structured reasoning process rather than a single model response. Visible expert disagreement was not perceived as a weakness but instead increased trust, as users could understand the underlying trade-offs and rationale behind the final decision. Scenario-specific reporting also proved critical: outputs tied directly to concrete system attributes, such as architecture design, dependency on external models, or missing safeguards, were significantly more persuasive than generalized safety statements. Finally, fallback execution modes did not materially reduce the perceived value of the system; as long as the evaluation structure, decision logic, and reporting format remained visible, the prototype effectively demonstrated how an institutional AI safety review process would function in practice.

Outcomes

The results indicate that the project successfully transformed a technically complex council architecture into a usable safety review workflow. The interface did not merely display outputs; it made the logic of the review process inspectable. This outcome matters because institutional AI governance depends on the ability to explain, not only to compute.

A second outcome is that the deployment layer became part of the governance story. By preserving council logic under constrained execution conditions, the prototype showed that evaluation workflows can be designed to fail gracefully rather than fail invisibly. This improves both presentation readiness and organizational confidence.

Implications

The theoretical implication of the project is that usability should be treated as a first-class component of AI governance implementation. The literature often frames governance as policy, testing, or documentation, but this project suggests that the interface layer is where those strands become actionable.

The practical implication is that institutions such as UNICC can improve the quality of AI review by making their evaluation process more structured, more visible, and easier to execute. A council architecture becomes significantly more valuable once its outputs are readable and its synthesis logic is transparent.

Summary

In summary, the proof of concept produced an evaluator-ready submission workflow, visible module independence, explicit council synthesis, and a deployable demonstration path. Together, these results confirm that the interface, testing, and deployment layer is a meaningful contribution to the Spring 2026 UNICC AI Safety Lab rather than a peripheral packaging exercise.

Repository of Data Sets and Code

Live Demo / Product Demonstration URL: http://unicc-ai-guard.replit.app

Public GitHub Repository: https://github.com/wesleykoe/UNICC

The repository contains the integrated Spring 2026 capstone product, including application code, council-of-experts logic, tests, and supporting documentation. Because this project focused on product workflow rather than on a separate public dataset, the most important shared artifacts are the source code, configuration structure, and testing materials used to demonstrate the end-to-end review process.

Issues Encountered

Several issues emerged during the project, most of which were anticipated by the risk planning process.

The first issue was scope coordination. Because the team product was intentionally split across platform research, expert-module logic, and interface/testing/deployment, the project could easily have drifted into duplicated effort or blurred responsibilities. This risk was mitigated through explicit role alignment and by defining my contribution around evaluator workflow, module display, testing logic, and deployment readiness rather than around re-implementing the entire model layer.

The second issue was infrastructure realism. A polished user experience is much harder to deliver when the backend may depend on heavy model loading, long-running inference, or unavailable credentials. Early versions of the workflow were vulnerable to looking unresponsive or incomplete when live execution was constrained. This issue was handled by designing a portability mode that preserved the architecture and output schema. The solution did not pretend the limitation did not exist; instead, it converted the constraint into a design requirement for graceful degradation.

The third issue was output consistency across modules. Multi-expert systems are only persuasive if their differences are substantive and stable. There was a recurring risk that module outputs might appear too similar, which would weaken both the evaluation score and the conceptual integrity of the system. This was addressed by keeping the display logic explicitly separated by expert role and by framing each module around different risk emphases.

The fourth issue was demonstration quality. Because the capstone involved both written submission and live product review, the team needed a workflow that could survive multiple audiences: faculty, technical evaluators, and non-technical stakeholders. This required repeated refinements to wording, layout, and explanation depth so that the final system could operate as a real review tool and as a defensible presentation artifact.

Risk management plan

These issues were closely tied to the risks identified earlier in the project. Scope ambiguity was mitigated through early alignment and repeated clarification of ownership boundaries. Infrastructure instability was mitigated through fallback execution and schema preservation. Output inconsistency was mitigated through role-specific presentation logic and structured council summaries. Demonstration risk was mitigated through iterative testing of the full reviewer journey rather than isolated page design.

None of the issues encountered ultimately exceeded the planned contingencies. In each case, the resolution strategy reflected a broader lesson: AI safety products should be designed not only for ideal execution but also for degraded execution, partial information, and institutional time pressure. The risk plan was therefore useful not simply as documentation but as a practical design guide.

Lessons Learned

The most important lesson from this project is that usability is a governance capability. Before this capstone, it would have been easy to think of interface work as downstream packaging. The project demonstrated the opposite. If users cannot submit, interpret, and escalate a review consistently, the organization does not really possess a functioning governance process, no matter how strong its theoretical framework may be.

A second lesson is that evaluator readiness should be designed intentionally. Products that are built only for internal familiarity often fail when placed in front of a first-time reviewer. The Spring 2026 evaluator guide made this insight explicit, but the product work reinforced it in practice. Dynamic input acceptance, visible module distinction, and readable synthesis all needed to be engineered deliberately.

A third lesson is the importance of graceful degradation. Institutional AI tools rarely run in perfect conditions. Designing for fallback mode, partial execution, and transparency about constraints is not a concession to weakness; it is part of producing a reliable system.

A fourth lesson concerns teamwork. A multi-component capstone only becomes coherent when each contributor is disciplined about boundaries and integration points. The strongest individual work in such a setting is not the work that expands endlessly in scope, but the work that strengthens the integrated final product.

Finally, I learned that product communication and technical communication are not separate skills in AI governance work. The best design decisions were often the ones that made the system easier to explain, easier to trust appropriately, and easier to document.

Conclusion and Further Work

Generative AI introduces significant limitations when applied to business process re-engineering, particularly in terms of robustness and generalizability. From a robustness perspective, model outputs are highly sensitive to prompt formulation, input variability, and execution conditions, which can lead to inconsistent or non-deterministic results across similar evaluation scenarios. This instability poses challenges for workflows that require repeatability and auditability, especially in institutional settings such as AI safety evaluation. From a generalizability perspective, generative models often perform well on structured or benchmark-aligned inputs but may struggle to maintain performance across diverse real-world contexts, including multilingual, domain-specific, or high-risk scenarios. As a result, systems built on generative AI may not fully transfer across operational environments without additional tuning and validation.

To address these limitations, future work should focus on improving both model stability and cross-context adaptability. This includes implementing more controlled evaluation pipelines, standardizing prompts and test cases, and integrating hybrid approaches that combine deterministic rules with generative reasoning. In addition, expanding testing coverage across multilingual and multimodal datasets will help improve generalizability, while incorporating human-in-the-loop validation can enhance robustness in high-stakes decision-making scenarios.

Conclusions

The project successfully addressed its primary objective of making the Spring 2026 UNICC AI Safety Lab more usable, testable, and evaluator-ready. The developed system demonstrates that the research question has been effectively answered, as it delivers a complete workflow capable of accepting dynamic inputs, presenting three distinct expert perspectives, and synthesizing a final council decision. The key finding from this work is that the effectiveness of an AI governance system depends not only on model capability or policy design, but critically on workflow structure and interpretability. Features such as dynamic submission, visible module independence, and transparent synthesis of disagreement proved essential in transforming the council architecture into a practical institutional review tool. From a viability standpoint, the project can be considered a successful proof of concept, as it delivers the core components required for evaluation, demonstration, and iteration, even though it does not attempt to meet full enterprise deployment requirements.

Implications

This project carries both theoretical and practical implications. From a theoretical perspective, it supports the argument that AI governance should be analyzed not only at the level of models or policy frameworks, but also at the level of workflow design and human interaction. Institutional trust in AI systems is shaped in part by how evaluation processes are structured, surfaced, and interpreted by users, extending existing research on transparency and accountability. From a practical standpoint, the project provides a clear and actionable blueprint for organizations such as UNICC to operationalize multi-expert AI safety evaluation. It demonstrates how complex governance logic can be translated into an accessible workflow that is usable by developers, reviewers, and decision-makers, while maintaining alignment with evaluation standards without reducing the system to superficial compliance.

Limitations

The project highlights several important limitations, particularly related to the use of generative AI in business process re-engineering. In terms of constraints, the prototype was developed within a semester-long capstone and therefore prioritizes evaluator workflow and demonstration capability over enterprise completeness, leaving out features such as authentication, long-term audit storage, and large-scale deployment infrastructure. More fundamentally, generative AI introduces limitations in robustness and generalizability. Model outputs remain sensitive to prompt design, input variation, and execution conditions, which can lead to inconsistencies in evaluation results. In addition, the system has been primarily tested on structured scenarios, meaning that its performance may not fully generalize to more complex, multilingual, or real-world operational environments. From a validity perspective, some evaluation pathways relied on constrained or fallback execution rather than unrestricted live model operation, which limits the ability to draw strong conclusions about production-level robustness and may introduce bias toward expected outcomes in controlled test settings.

Further Work

Future work should focus on addressing these robustness and generalizability limitations while expanding the system toward production readiness. In the short term, this includes integrating the system with production-grade execution environments such as DGX infrastructure, expanding benchmark coverage to include multilingual and multimodal use cases, and improving consistency through standardized prompts and evaluation protocols. In the longer term, the system can evolve into a persistent institutional control layer for AI governance, incorporating human-in-the-loop escalation, reusable and versioned test packs, historical comparison across model iterations, and tighter integration between policy updates and evaluation logic. Addressing generalizability will require broader validation across domains and the development of adaptive evaluation frameworks capable of handling diverse real-world inputs.

Closing Summary

This project contributes a professionally documented and practically validated model for operationalizing a council-of-experts AI safety system. By focusing on workflow design, interpretability, and usability, it helps bridge the gap between high-level AI governance principles and the practical process of determining whether an AI system is safe to deploy. The results demonstrate that effective AI governance requires not only strong models and policies, but also structured, transparent, and user-centered evaluation workflows.

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Appendix A — Project Acceptance Document

Sponsor's Project Acceptance Document

This document is the means by which your project sponsor formally agrees that your project has been satisfactorily completed and that it meets the project goal and objectives that were set at the onset of the project. It is therefore important that you describe the goal, objectives, and related metrics in the appropriate section below. The "PLAN" section is to be prepared at the beginning of the project and the "RESULTS" and "ACCEPTANCE" sections after your project has been completed. Your sponsor should provide input and sign where indicated. The signed document will also be a required section in your Project Final Report. This document is a template whose sections may be expanded as necessary.

PLAN

(To be filled out and signed at the **start** of the project)

Project Name: Operational AI Safety Lab: Council-of-Experts Architecture for the UNICC AI Sandbox

Student Name: Yuze Wu

Sponsoring Organization: UNICC & New York University

Project Sponsor Name and Title: Dr. Andrés Fortino, Clinical Associate Professor

Project Sponsor Contact Information (email and phone): agf249@nyu.edu

Planned Start Date: 27/01/2026 **Planned End Date: 18/04/2026**

PROJECT PLAN

At the project start, show the project goal; the project objectives, and related metrics to be used to show successful project completion. The sponsor should sign to indicate agreement. Please use PDF version for signatures

Project Goal

To design, develop, and deploy a functional AI Safety Lab prototype that integrates multiple AI safety evaluation modules into a Mixture-of-Experts (MoE) Council architecture capable of evaluating AI agents prior to deployment in the UNICC AI Sandbox.

Objective #1
Design the system architecture and define a standardized JSON schema for expert module outputs and arbitration logic.

Metric:

- **Architecture diagram completed**
- **Unified schema validated across all expert modules**
- **Faculty sponsor approval by March 1, 2026**

Objective #2
Develop and implement the Council-of-Experts arbitration mechanism integrating three independent safety evaluation modules.

Metric:

- **All expert modules return structured outputs**
- **Ensemble logic successfully synthesizes final evaluation**

≥ 95% functional testing success rate

Objective #3

Build and deploy a web-based testing interface enabling AI agent evaluation and risk-tier classification.

Metric:

- **Interface supports structured input**
- **Risk-tier tagging system operational**
- **No critical system errors during demo testing**

Objective #4
Validate the system through controlled testing scenarios and prepare deployment-ready demonstration for NYU AI Sandbox.

Metric:

- **Completion of at least 20 structured test cases**
- **Consistent evaluation behavior across scenarios**
- **Successful live demonstration to sponsor**

I agree with the above-planned project goal, project objectives, and related metrics.

Andres Fortino
______________________________ ________________
Project Sponsor Signature **Date**

Appendix B — Project Sponsor Agreement

New York University
MS in Management and Systems
Applied Project
Project Sponsor Agreement

Goals of the Program

For Participating Organizations

- Begin relationship with New York University
- Receive help from highly trained NYU graduate student
- Provide internship opportunity for NYU graduate student
- Receive assistance at no cost

For NYU Graduate Students

- Manage and implement a meaningful project aligned with their professional and educational goals
- Hands-on experience interacting with a start-up or operational small business or organization
- Earn credit toward completion of graduate degree by conducting an unpaid Applied Project under the mentorship of an NYU-SCPS professor.

Project Sponsor and Student Responsibilities

- Student prepares project planning documents
- Sponsor reviews and approves student's project plan
- Student submits project plan to faculty supervisors for approval
- Student conducts project according to plan
- At predetermined milestones sponsor reviews and approves status reports submitted by student
- Status reports reviewed and evaluated by faculty supervisors to assure student effort and project meet course requirements
- Project sponsor and student participate in periodic project reviews with NYU
- At project completion project sponsor completes evaluation forms
- Student prepares final report

Project Selection Process

- Project Evaluation Committee reviews proposed projects
- Projects are:
  - Relevant to MS degree course content
  - Significant to the participating organization
  - Substantial in terms of duration and scope
  - Challenging to the student
  - Capable of being measured against predetermined goals

The MS in Management and Systems

Concentrations in:

- Strategy and Leadership
- Systems Management
- Database Technologies
- Enterprise Risk Management

Typical Participating Student Profile

- Students selected to participate in this program meet stringent criteria
- Have completed all coursework
- High achievers with highest level GPAs and strong academic credentials
- 2-10 years of business experience

- Highly motivated for success

Sponsor and Project Information

Type of Organization	☐ For Profit ☑ Not for Profit				
Name of Organization	United Nations International Computing Centre (UNICC)				
Address	7 East 12th Street				
City	New York	State	NY	Zip	10003
Project Sponsor	First Name	Andrés	Last Name	Fortino	
Title	Clinical Associate Professor				
Phone	SPS Office				
Email	agf249@nyu.edu				
Web Site	https://www.sps.nyu.edu				
Type of Business	Higher Education / AI Governance Research / International Public Sector Collaboration				

Student Name	Yuze Wu
Project Title	Operational AI Safety Lab: Multi-Module Council-of-Experts Architecture for the UNICC AI Sandbox

Description of Project	
This project develops an operational AI Safety Lab prototype based on a standalone Small Language Model (SLM). The system integrates three independent AI safety testing agents into a Mixture-of-Experts (MoE) “Council of Experts” architecture. Each expert module evaluates AI agents across governance-aligned dimensions such as fairness, robustness, transparency, and compliance risk. The final system includes a testing interface, risk-tier classification mechanism, ensemble arbitration logic, and deployment readiness within the NYU AI Sandbox environment.	
Estimated Hours of Student Participation	250-270 hours

Anticipated Results
Delivery of a fully functional AI Safety Lab prototype capable of evaluating AI agents prior to deployment in the UNICC AI Sandbox. The final product will include: ● Multi-module inference ensemble (MoE architecture)

• Council-of-Experts arbitration logic • Risk-tier tagging system • Web-based testing interface • Deployment-ready prototype on NYU infrastructure

Knowledge and expertise student will need to be able to complete the project
• AI governance and risk management principles • LLM evaluation and prompt engineering • Mixture-of-Experts (MoE) system design • Python backend development • Web interface development • JSON schema design and evaluation metrics • AI safety testing methodologies

Will the project sponsor be available for periodic meetings with NYU to review progress, address questions and concerns with the professor supervising the program? *This is a requirement for the program*	☑ Yes ☐ No
Describe the form and frequency of supervision of the student by the Project Sponsor. Weekly or bi-weekly progress meetings with faculty sponsor. Formal milestone reviews at key development phases (design, implementation, validation, final delivery). Additional consultations as needed for architecture, governance alignment, and deployment strategy.	

Sponsor Agreement

Students are interns, not professional consultants. NYU is not responsible for the outcomes of projects undertaken by students. Work is on a best-efforts basis; no guarantees or warranties are expressed or implied. Organization is responsible for evaluating work presented, determining its value and whether to use it or not. Some projects may require on-going management or even re-work by the Organization after the student completes their Applied Project.

Please note that in order to post an unpaid position, the internship must encompass all 6 components below:

1. The internship, even though it includes actual operation of the facilities of the employer, is similar to training which would be given in an educational environment;
2. The internship experience is for the benefit of the intern;
3. The intern does not displace regular employees, but works under close supervision of existing staff;
4. The employer that provides the training derives no immediate advantage from the activities of the intern; and on occasion its operations may actually be impeded;
5. The intern is not necessarily entitled to a job at the conclusion of the internship; and
6. The employer and the intern understand that the intern is not entitled to wages for the time spent in the internship.

I have read and agree with the information shown in the Terms and Conditions for employers contained on the following web page(s): http://www.nyu.edu/life/resources-and-services/career-development/employers/post-a-job/terms-and-conditions.html

Please complete and sign this form in the space provided below and return to the course professor via the student who will upload the document to the course drop-box. For any questions, please email the professor: Prof. Israel Moskowitz im36@nyu.edu.

I agree to all of the above

Participating Organization New York University Date ___________

By (signature): Andres Fortino
Project Sponsor

Printed Name: Andrés Fortino

Title: Clinical Associate Professor / NYU School of Professional Studies

Student Agreement

Students who are planning to conduct an unpaid Applied Project must read and agree to the "Important Considerations Before Accepting a Job or Internship" contained on the following web page(s): http://www.nyu.edu/life/resources-and-services/career-development/find-a-job-or-internship/important-considerations-before-accepting-a-job-or-internship.html.

Students do not register their Applied Project with the Wasserman Center.

I agree to the all of the above

Student Name (Print) Yuze Wu Date 02/20/2026

Signature: Yuze Wu

Appendix C — Functional Requirements Specifications

MASTER OF SCIENCE IN MANAGEMENT AND SYSTEMS

Applied Project Capstone

MASY GC-4100

UNICC AI Safety Lab — Council of Experts MoE Agent

Project Goal

The goal of this project is to design and implement a Council-of-Experts AI Safety Agent using a Mixture-of-Experts (MoE) architecture that integrates three independently developed AI Safety Agents from the Fall 2025 UNICC competition into a unified inference ensemble.

The system will be deployed as a proof-of-concept within the NYU AI Sandbox environment, demonstrating that AI governance, risk assessment, and trust & safety validation can be operationalized in a standalone Small Language Model (SLM) environment.

The final system will provide:

- Multi-agent safety evaluation
- Veto-weighted consensus decision logic
- Structured JSON audit outputs
- Risk-tier classification (T1–T3)
- Transparent and reproducible evaluation results

This supports UNICC's objective to evaluate AI agents before production deployment in the UN ecosystem.

Project Objectives

Objective 1 — Operationalize Multi-Agent Evaluation

Port or re-implement the top three Fall 2025 safety agents as independent expert modules. Each expert must:

- Produce structured outputs
- Evaluate AI agents independently
- Follow defined safety principles (Fairness, Robustness, Transparency, Security, etc.)

Objective 2 — Design Council Decision Logic

Develop a Veto-Weighted MoE Council Architecture that:

- Aggregates expert outputs
- Applies weighted consensus logic
- Enables minority veto override for high-risk flags
- Produces final structured CouncilDecision output

Objective 3 — Deploy on Sandbox-Compatible Environment

The system must:

- Be portable to NYU DGX Spark cluster
- Run without reliance on external APIs
- Operate using open-source SLM weights (future integration)
- Produce auditable outputs for governance review

Objective 4 — Provide Human-Usable Interface

Deliver:

- Web-based interface (API & text-based testing modes)
- Structured risk-tier tagging
- Visualized evaluation outputs

- Clear audit trail for UNICC IT developers (intended end users)

Requirements Specifications

1. Purpose and Scope

The AI Safety Lab functions as a pre-deployment AI evaluation gate within the UNICC AI Sandbox.

Target Users:

- UNICC IT developers
- AI product owners within the UN system

Scope:

- Safety evaluation of AI agents
- Governance and risk scoring
- Structured safety certification support

Out of Scope:

- Training foundational LLMs
- Live production deployment in UN systems
- Replacing human oversight

2. User Interaction

The system must support:

Input Methods

- API-based model endpoint testing
- Text-based evaluation mode (fallback when APIs unavailable)

Supported Formats

- JSON structured responses

- Safety scorecards
- Risk tier classification

3. Response Generation

Each expert module must:

- Accept AI agent outputs
- Evaluate against safety principles
- Return structured output

Example expert output:


```
{   "expert_id": "policy_expert",   "risk_score": 0.67,   "risk_tier":
"T2",   "confidence": 0.82,   "veto": false,   "justification":
"Detected borderline fairness issue." }
```


The Council layer must:

- Aggregate expert responses
- Apply veto-weighted logic
- Output final decision

Example council output:


```
{   "final_risk_tier": "T2",   "consensus_score": 0.71,
"veto_triggered": false,   "explanation": "2/3 experts classified as
moderate risk." }
```


4. Learning and Adaptation

This PoC does not include real-time model training. However, the architecture must support:

- Incremental improvement
- Updated expert modules
- Modular replacement of safety evaluators

5. Integration with Other Systems

The system must:

- Integrate the three Fall 2025 agents

- Support integration with the AI Safety Framework module, the Safety Agent backend module, and the Testing Interface module
- Produce auditable outputs compatible with governance documentation

6. Performance Requirements

- Response time < 5 seconds per evaluation (PoC target)
- Deterministic risk-tier classification
- 100% structured JSON output validity
- Stable execution without critical failures

7. User Interface Design

The UI must support dual testing modes and display:

- Accuracy / Precision / Recall
- Principle-level scores
- Confusion matrix
- Risk tier (T1 / T2 / T3)
- Highlighted failure cases
- Council consensus breakdown

8. Development Platform

Development stack:

- Backend: Python (FastAPI)
- Council Logic: Custom MoE arbitration module
- Frontend: Replit-hosted Web UI (HTML/JS)
- Data format: JSON
- Deployment target: NYU DGX Spark cluster

Use Cases

Use Case 1 — AI Chatbot Safety Evaluation

- UNICC developer submits AI chatbot endpoint
- System sends test prompts
- Three expert modules independently evaluate
- Council aggregates outputs
- Final risk tier assigned
- Structured report generated
- Developer receives safety decision before deployment

Use Case 2 — Text-Based Offline Evaluation

- User pastes model output
- Expert modules analyze response
- Council logic computes final risk
- System returns risk tier and justification
- Output saved for audit

Documentation of LLM Prompts

How ChatGPT Was Used

ChatGPT was used as a structured drafting assistant to help transform conceptual system architecture into a formal Functional Requirements Specification (FRS) document aligned with the MASY GC-4100 Memorandum template. Specifically, the model was used to:

- Translate high-level system design concepts into structured requirement statements
- Organize functional components according to the required FRS dimensions (Purpose, Interaction, Integration, Performance, etc.)
- Generate example structured JSON outputs to clarify expected system behavior
- Refine wording for clarity, precision, and alignment with academic standards

The model was not used to invent system functionality. All architectural decisions, logic design, and technical specifications were defined by the author prior to drafting.

Prompt Examples

The following example prompts illustrate how the model was used during drafting:

Example Prompt 1 — Structuring Requirements

"Convert the following system description into a formal Functional Requirements Specification section. Ensure alignment with Purpose, Scope, User Interaction, and Performance categories."

Example Prompt 2 — Clarifying Technical Outputs

"Provide an example of structured JSON output for a multi-expert risk evaluation system that includes risk tier, score, confidence, veto flag, and justification."

Example Prompt 3 — Refining Academic Tone

"Rewrite the following paragraph in a formal academic style appropriate for a Capstone Project Requirements Memorandum."

These prompts were iterative and adjusted to ensure outputs remained aligned with the intended system architecture.

Iterative Refinement Process

The drafting process followed a structured refinement workflow:

- Initial manual outline of system architecture and core components
- Prompting ChatGPT to organize content into FRS-compliant sections
- Reviewing generated text for technical accuracy and removing ambiguities
- Revising prompts to improve specificity and reduce overgeneralization

- Cross-checking all requirements against actual system capabilities

Multiple refinement cycles were conducted to ensure that requirements were measurable and specific, functional statements reflected implemented or planned features, and terminology remained consistent across sections. The final document reflects human-edited and validated content.

Validation Method

To ensure accuracy and authenticity:

- All functional requirements were cross-validated against the implemented MVP prototype
- JSON structures included in this document were tested in the working system
- Performance requirements were aligned with actual system constraints
- Architectural claims were verified against the designed Council-of-Experts logic

The use of ChatGPT was limited to drafting support. Final technical validation, requirement definition, and architectural decisions were performed independently by the author.

Conclusion

This Project Requirements Specification defines the architecture, objectives, and implementation plan for the UNICC AI Safety Lab Council-of-Experts MoE Agent.

The solution aligns with UNICC AI Sandbox governance goals, the intended UNICC IT developer end users, and the integration of Fall 2025 expert models. The final product demonstrates that AI governance can be operationalized through modular, auditable, and scalable multi-agent ensemble systems suitable for institutional environments.

Prototype Demonstration (MVP)

To validate the feasibility of the proposed Council-of-Experts MoE architecture, a working MVP prototype has been implemented and deployed in a sandbox environment.

Live Demo: https://attached-assets-1-yw8512.replit.app

The MVP demonstrates:

- Independent expert module outputs
- Structured JSON risk evaluation
- Tier classification (T1–T3)
- Consensus-based council decision logic
- Preliminary veto-weighted aggregation mechanism

The interface simulates integration of the Team 6 Technical Agent, Team 8 Governance Workflow, and Team 1 Benchmark Framework.

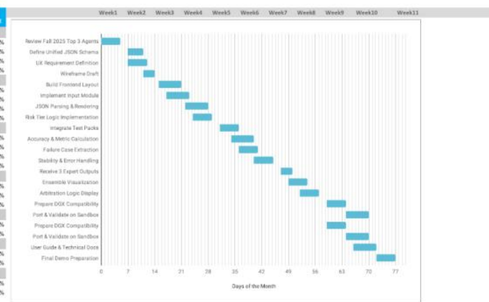
Appendix D — Project Plan

The project baseline list of tasks and milestones for the AI Safety Lab — Project 3 Testing & Deployment Timeline is shown below. The original document is preserved in landscape orientation for readability.

AI Safety Lab – Project 3 Testing & Deployment Timeline

* = an automatically calculated cell

TASK NAME	START DATE	END DATE	START ON DAY*	DURATION* (WORK DAYS)	TEAM MEMBER	PERCENT COMPLETE
Requirements & Planning						
Review Fall 2025 Top 3 Agents	1/27	2/2	0	5	Dennis	100%
Define Unified JSON Schema	2/3	2/6	7	4	Dennis	100%
UX Requirement Definition	2/3	2/7	7	5	Dennis	100%
Wireframe Draft	2/7	2/10	11	3	Dennis	100%
Interface Development						
Build Frontend Layout	2/11	2/18	15	6	Dennis	100%
Implement Input Module	2/13	2/20	17	6	Dennis	100%
JSON Parsing & Rendering	2/18	2/25	22	6	Dennis	100%
Risk Tier Logic Implementation	2/20	2/26	24	5	Dennis	100%
Testing Module						
Integrate Test Packs	2/27	3/5	31	5	Dennis	100%
Accuracy & Metric Calculation	3/2	3/9	34	6	Dennis	100%
Failure Case Extraction	3/4	3/10	36	5	Dennis	100%
Stability & Error Handling	3/8	3/14	40	5	Dennis	100%
Council of Experts Integration						
Receive 3 Expert Outputs	3/15	3/18	47	3	Dennis	80%
Ensemble Visualization	3/17	3/23	49	5	Dennis	80%
Arbitration Logic Display	3/20	3/26	52	5	Dennis	60%
Council of Experts Integration						
Prepare DGX Compatibility	3/27	4/2	59	5	Dennis	20%
Port & Validate on Sandbox	4/1	4/8	64	6	Dennis	0%
Sandbox Porting						
Prepare DGX Compatibility	3/27	4/2	59	5	Dennis	5%
Port & Validate on Sandbox	4/1	4/8	64	6	Dennis	0%
Documentation & Final Prep						
User Guide & Technical Docs	4/3	4/10	66	6	Dennis	5%
Final Demo Preparation	4/9	4/15	72	5	Dennis	0%



The project plan was developed at the start of the Spring 2026 capstone and used as the baseline against which actual progress was tracked. Tasks were grouped into five phases: Requirements & Planning, Interface Development, Testing Module, Council of Experts Integration, Sandbox Porting, and Documentation & Final Prep. Most tasks completed on or near their planned dates; the Sandbox Porting and Final Demo Preparation phases experienced minor adjustments as DGX availability became clearer late in the term, but no critical milestones were missed.

Appendix E — Risk Management Plan

MASTER OF SCIENCE IN MANAGEMENT AND SYSTEMS

Applied Project Capstone

MASY GC-4100

MEMORANDUM

This project aims to develop a proof-of-concept AI Safety Lab for the UNICC AI Sandbox by integrating three independent AI safety agents into a unified "Council of Experts" architecture. The system evaluates AI tools for governance, threat, and behavioral risks before deployment, providing structured and auditable safety decisions.

Risks

Number	Risk	Probability (1–3)	Impact (1–3)
1	Integration of three expert modules fails due to incompatible architectures or outputs	3	3
2	API / backend instability (FastAPI, ngrok, deployment issues) affects system reliability	3	2
3	Model inference latency too high (10+ min) causing poor demo experience	2	3
4	Lack of evaluation metrics / model validation reduces credibility of results	2	3
5	UI is not user-friendly for UN stakeholders (too complex or unclear)	2	3
6	Team coordination issues lead to delays in integration	2	2
7	Scope creep (adding unnecessary features like extra dashboards or modes)	2	2
8	Failure to deploy or demonstrate in sandbox / live environment	3	3

Number	Risk	Probability (1–3)	Impact (1–3)
9	Incomplete documentation (model cards, explanation, slides) affects grading	2	2
10	Adapter / model loading issues (path, config, compatibility) break pipeline	3	2

Risk Matrix

Each risk is plotted in the matrix below by probability of occurrence (rows) and impact / exposure (columns).

	1. Slight Impact	2. Moderate Impact	3. High Impact
1. Very Unlikely			
2. Possible		6, 7, 9	3, 4, 5
3. Expected		2, 10	1, 8

Contingency Plan

#	Description	Prob.	Exp.	Contingency Plan
1	Integration of three expert modules fails due to incompatible architectures	3	3	Standardize output format using JSON schema and implement an arbitration layer. If integration fails, fallback to a simplified single-expert demo with simulated council output.
2	Backend/API instability (FastAPI, ngrok, connection issues)	3	2	Maintain both local and ngrok deployment versions. Pre-test endpoints and enable mock response mode if API fails during demo.
3	Model inference latency too high for real-time demo	2	3	Pre-run evaluations and cache results. Reduce test cases during demo and avoid real-time inference when presenting.
4	Lack of evaluation metrics reduces	2	3	Introduce basic evaluation metrics such as risk counts and severity levels. Use simulated

#	Description	Prob.	Exp.	Contingency Plan
	credibility			benchmark cases if real evaluation is unavailable.
5	UI/UX not user-friendly for stakeholders	2	3	Simplify interface by focusing on key outputs (Council Decision, risks). Remove unnecessary features and improve visual clarity.
6	Team coordination issues causing delays	2	2	Hold weekly check-ins with clearly assigned ownership areas. Use a shared task tracker so blockers surface early.
7	Scope creep from adding unnecessary features	2	2	Maintain a documented scope baseline and require explicit approval before adding features. Defer non-essential ideas to future-work notes.
8	Failure to deploy or demonstrate system in sandbox	3	3	Prepare backup demo including precomputed results and recorded demo video. Use multiple deployment options (local, ngrok, Replit).
9	Incomplete documentation affecting grading	2	2	Treat documentation as a working artifact updated weekly. Conduct a pre-submission review against the rubric and template.
10	Adapter/model loading issues breaking pipeline	3	2	Validate all model paths and configs in advance. Keep backup checkpoints and fallback to precomputed outputs if loading fails.

Appendix F — Organizational Change Management Plan

Organizational Change Plan

The Organizational Change Plan

Problem and Selected Solution

The current AI evaluation process within UNICC relies heavily on manual, analyst-driven reviews, which are time-consuming, inconsistent, and difficult to scale as AI adoption increases. This creates operational inefficiencies and exposes the organization to governance, compliance, and reputational risks.

Alternative solutions considered included maintaining the manual process with improved documentation, adopting a fully ML-based automated evaluation system, and implementing a hybrid safety testing system. The manual option did not solve scalability problems, while the ML-only option lacked interpretability and auditability. Therefore, the hybrid AI Safety Testing System was selected because it combines rule-based governance logic with automated evaluation and produces structured, reviewable outputs.

Objective Clarification

Objectives to be achieved

The main objective is to implement a standardized AI Safety Testing System that improves the speed, consistency, and transparency of AI evaluations within UNICC. Using the SMART approach, the project aims to reduce evaluation time during the pilot phase, improve consistency across review outcomes, and provide audit-ready reports that support organizational governance goals.

Vision

The technological application aims to create a reliable and scalable AI safety evaluation process that supports responsible AI adoption within UNICC. It aligns with the organization's strategic goals by strengthening governance, reducing operational risk, and enabling safer deployment of AI tools in the UNICC AI Sandbox.

Key Outcomes

Success will be demonstrated through a pilot MVP in the UNICC AI Sandbox that shows faster evaluation cycles, more consistent results, and clearer reporting. Key metrics will include reduction in processing time, user adoption during the pilot, output consistency across repeated evaluations, and stakeholder satisfaction with the system's usability and transparency.

Stakeholder Engagement

Identify Stakeholders

Key stakeholders include UNICC developers as primary users, governance and compliance teams as reviewers, non-technical evaluators who need understandable outputs, and leadership who will approve broader adoption. These stakeholders will be identified, prioritized by influence and interest, and assessed based on their needs, expectations, and possible concerns about workflow changes.

A stakeholder engagement plan will be developed to address different levels of involvement and resistance. High-influence stakeholders will be engaged early in decision-making, while end users will be involved through pilot participation, feedback collection, and training sessions.

Communication Plan

Communication will be delivered through workshops, status updates, demonstration sessions, and written summaries tailored to different stakeholder groups. Messages will focus on efficiency gains, improved compliance, and the role of the system as a support tool rather than a replacement for human judgment, with communication occurring weekly during the pilot and bi-weekly during rollout.

Cultural Assessment and Planning

Current Culture Analysis

The current organizational culture values careful review, human judgment, and compliance, but it is still heavily dependent on manual processes. While this creates caution and accountability, it also limits scalability and may create resistance toward automation because of concerns about trust, accuracy, and role changes.

Desired Culture Definition

The desired culture is one that supports responsible innovation, structured decision-making, and collaboration between technical and non-technical teams. The implementation aims to foster transparency, adaptability, and trust in standardized AI-supported evaluation processes while preserving appropriate human oversight.

Change Management

Change Model Adoption

Kotter's 8-Step Change Model is the most appropriate framework for this implementation because it supports both urgency and gradual adoption. It helps the organization build awareness of the limitations of the current process, create a guiding coalition, communicate the value of the new system, and reinforce adoption through visible pilot results.

Resistance Management

Potential resistance may come from staff who worry that automation will reduce the value of their work or produce unreliable outputs. Resistance will be managed by identifying concerns early, involving users in the pilot, clearly explaining the human-in-the-loop approach, and showing how the system improves consistency rather than replacing professional judgment.

Skills and Capability Enhancement

Skills Inventory

UNICC currently has strong domain knowledge in governance and AI evaluation, but there are capability gaps in the use of automated safety testing tools, structured output interpretation, and system-supported review workflows. These gaps will be addressed through targeted training and, if needed, limited talent support in AI governance or technical integration.

Training and Development

Training will cover system navigation, interpretation of test outputs, risk-tier classification, and escalation procedures when results need human review. Refresher training and onboarding materials will also be created so that new staff can adopt the system consistently after the pilot phase.

Documentation and Communication

Process Documentation

Every stage of the project will be documented, including planning decisions, pilot setup, testing procedures, issues encountered, and lessons learned. The technology trial will include usability testing, workflow validation, repeated evaluation checks, and reporting reviews to ensure that the system functions reliably and produces meaningful outputs.

Feedback Mechanisms

Feedback will be collected through user surveys, pilot debrief sessions, stakeholder review meetings, and system usage logs. This feedback will be reviewed regularly so that the strategy, workflows, and interface can be adjusted before full-scale rollout.

Integration and Adaptation

Scalability Plan

The technology will first be implemented as a pilot in the UNICC AI Sandbox and then expanded gradually into broader standard operations if the pilot meets performance and adoption targets. Scaling will include integration into normal review workflows, additional user onboarding, and phased expansion across relevant teams.

Continuous Improvement

A continuous improvement process based on Six Sigma principles will be used to monitor errors, reduce inefficiencies, and improve the system over time. Regular reviews of metrics, user feedback, and process performance will help ensure that the technology continues to support both operational needs and organizational change.

Appendix G — Technology Trial Plan

MASTER OF SCIENCE IN MANAGEMENT AND SYSTEMS

Applied Project Capstone

MASY GC-4100

Technology Trial Plan — UNICC

Business Task and Objectives

The United Nations International Computing Centre (UNICC) is developing tools to evaluate the safety and compliance of AI systems deployed across UN agencies. Currently, AI safety reviews in the UNICC AI Sandbox rely heavily on manual analyst evaluation. This process is time-consuming, inconsistent, and difficult to scale as the number of AI systems grows.

The objective of this technology trial is to evaluate whether an automated AI Safety Evaluation Agent can improve the efficiency, consistency, and reliability of AI safety assessments compared with traditional manual review.

Specifically, the trial will test whether automated safety evaluation tools can:

- Reduce evaluation time
- Improve consistency across evaluations
- Maintain or improve safety detection accuracy
- Provide structured and reproducible safety reports

The results will help determine whether automated AI safety evaluation tools should be adopted within the UNICC AI Sandbox workflow.

The Hypothesis

The hypothesis for the technology trial is:

H1:

Using an automated AI Safety Evaluation Agent will significantly improve the efficiency and consistency of AI safety evaluations compared with manual analyst reviews.

More specifically:

- H1a: Automated evaluation will reduce the time required to complete safety assessments.
- H1b: Automated evaluation will produce more consistent safety scores across multiple runs.
- H1c: Automated evaluation will identify safety issues (e.g., bias, toxicity, hallucination) with equal or greater accuracy than manual review.

Null Hypothesis (H0):

The automated AI Safety Evaluation Agent will not significantly improve efficiency, consistency, or detection accuracy compared with manual safety evaluation methods.

Baseline Metrics and Control Group

The baseline for this trial will be the current manual AI safety evaluation process used by analysts reviewing AI outputs in the UNICC AI Sandbox.

Control Group (Group A — Manual Evaluation)

In the control condition, human reviewers will evaluate AI outputs manually using standard evaluation guidelines. Reviewers will analyze AI responses for bias, toxicity, hallucination, and policy compliance, and will record their evaluation results using a standard evaluation form.

Baseline Metrics

The following baseline metrics will be measured for the control group:

- Evaluation Time — average time required to complete a safety review

- Evaluation Consistency — variance in safety scores across multiple evaluators
- Issue Detection Rate — percentage of safety issues correctly identified
- Reviewer Agreement — degree of agreement between multiple reviewers

These baseline metrics will serve as the reference point for evaluating the automated system.

The Trial

The trial will use an A/B testing design comparing manual safety evaluation (Group A) with automated safety evaluation (Group B).

Group A — Manual Evaluation

Participants in the control group will perform traditional safety reviews using manual analysis. Steps include reviewing AI-generated outputs, identifying potential safety risks, recording findings manually, and assigning safety scores based on guidelines.

Group B — Automated Evaluation

The experimental group will use the AI Safety Evaluation Agent. Steps include submitting the same AI outputs to the automated system, running safety tests covering toxicity detection, bias detection, hallucination detection, and policy compliance checks, and generating a structured safety scorecard.

Trial Duration

The trial will run for two weeks, allowing sufficient data collection across multiple evaluation tasks.

Test Dataset

The trial will evaluate approximately 100 AI-generated outputs covering different safety scenarios, including harmful content, biased responses, factual hallucinations, and policy violations. Both groups will evaluate the same dataset.

The Technology

The technology tested in this trial is the AI Safety Evaluation Agent, a prototype system designed to automate AI safety assessments. The system integrates several components.

1. Policy Rule Engine

A deterministic rule system that converts governance standards into executable safety checks.

2. Machine Learning Safety Detectors

ML models are used to detect:

- Toxic language
- Bias in responses
- Hallucinated information
- Jailbreak vulnerabilities

3. Evaluation Pipeline

The pipeline performs the following steps: input AI response, run policy rule checks, run ML safety detectors, and generate a safety scorecard.

4. Structured Output

The system produces standardized outputs including safety scores, risk category, explanation of detected issues, and an evaluation summary. This structure allows easy comparison with manual evaluations.

Data Collection

Data will be collected during the trial from both manual and automated evaluations.

Quantitative Data

- Evaluation completion time

- Number of safety issues detected
- Safety score outputs
- Accuracy of issue detection

Qualitative Data

- Reviewer feedback on system usability
- Perceived reliability of automated results
- Clarity of evaluation reports

All evaluation results will be stored in structured datasets to enable statistical analysis.

Analysis of Results

The collected data will be analyzed to compare the performance of manual and automated safety evaluations.

Efficiency Analysis

Average evaluation time will be compared between the two groups. Metrics include mean evaluation time and percentage reduction in time.

Accuracy Analysis

Detection accuracy will be evaluated by comparing identified issues against a predefined ground truth dataset. Metrics include precision, recall, and F1-score.

Consistency Analysis

Consistency will be measured by comparing variance in evaluation scores. Lower variance indicates higher consistency.

Statistical Testing

Statistical tests (e.g., t-tests) may be used to determine whether observed differences are statistically significant.

Findings and Recommendations

Based on the results of the trial, the following outcomes are expected. If the automated AI Safety Evaluation Agent demonstrates faster evaluation times, equal or higher accuracy, and more consistent results, then the technology should be recommended for integration into the UNICC AI Sandbox workflow.

Adopting automated AI safety testing could:

- Reduce manual workload for analysts
- Improve consistency of safety evaluations
- Enable scalable safety governance for AI systems

However, if the system shows limitations in detection accuracy or usability, further refinement will be recommended before deployment. Future improvements may include expanding safety test coverage, improving model accuracy, and integrating real-world AI deployment scenarios.

Documentation of LLM Use

ChatGPT was used to assist in drafting the technology trial design, structuring the memorandum, and aligning the trial plan with the assignment rubric. The tool helped generate structured explanations for the trial methodology, evaluation metrics, and analysis framework.

Prompt Examples

Example prompts included:

- “Design an A/B technology trial for testing an AI safety evaluation system.”
- “Explain how to measure efficiency and accuracy in an AI safety evaluation trial.”
- “Structure a technology trial memorandum following academic rubric criteria.”

Appendix H — Status Reports

[Status Report A]

Project Status Report

Your Name: Yuze Wu

Project Title: UNICC

Date of report: 8 th March

1. Project Status and Explanation:

Project Status Area	Status (RYG)	Explanation
1. Overall Project Status	🟢	Project is progressing smoothly. Core architecture (Council of Experts framework) has been designed and partially implemented. Initial prototype is functional and aligns with project objectives.
2. Project Schedule	🟡	Slight delay (2–3 days) due to integration challenges between SLM and Gemini API. Timeline still manageable and does not affect final delivery.
3. Project Deliverables	🟢	Key deliverables including system design, expert framework, and initial Replit prototype have been completed. Remaining components are on track.
4. Resources & Collaboration	🟢	Team collaboration is effective. External resources (GitHub repo, Gemini API) are available and working. No resource constraints at this stage.
5. Changes	🟡	Adjusted system design to incorporate Gemini API for enhanced reasoning instead of relying solely on SLM. This improves performance but required minor rework.
6. Communication	🟢	Regular communication with team members and alignment with project direction. No communication issues.

For status above, indicate **Red**, **Orange**, or **Green**:

- **Red**: Critical issues, serious risks to project, significant intervention must occur to achieve success, potential for stoppage of project activity. Project slipping by 5+ days, and resources uncommitted to meet deliverables
- **Orange**: Some major issues, moderate risk to project, must monitor closely, some internal or/and external dissatisfaction with progress. Project plan slipping by 2+ days.
- **Green**: No major issues, minimal risk to project, on target with expected outcomes, project on schedule, everyone satisfied with progress.

2. **List All Completed Project Tasks:**
   - Designed overall system architecture (Council of Experts framework)
   - Defined expert roles (Bias, Safety, Hallucination)
   - Built initial Replit web interface (frontend + basic backend)
   - Set up Gemini API key and initial API connection
   - Developed prototype logic for expert-based evaluation workflow
   - Reviewed and integrated reference materials from Team 1, Team 6, and Team 8
   - Defined JSON output structure for evaluation scorecard

3. **List any concerns or issues that need the professor's involvement:**
   - Clarification on expected depth of technical implementation vs. proof-of-concept level
   - Guidance on evaluation metrics (how detailed scoring needs to be for final grading)
   - Confirmation on whether external APIs (e.g., Gemini) are fully acceptable for final submission

4. **Next series of tasks to complete:**
   - Convert expert modules into callable APIs (each expert as a service)
   - Complete integration between SLM and Gemini (hybrid evaluation logic)
   - Implement aggregation layer for combining expert outputs
   - Finalize JSON scorecard format and visualization
   - Conduct test cases (bias, toxicity, hallucination scenarios)
   - Optimize UI for better usability (Replit frontend improvements)
   - Prepare final report and presentation materials

5. Sponsor Signoff

Sponsor indicates agreement with the above status report:

By (signature): Andrea Fortino
Project Sponsor

Printed Name: Andrés Fortino
Please print in English

[Status Report B]

Project Status Report

Your Name: Yuze Wu

Project Title: UNICC

Date of report: 3rd Of April

1. Project Status and Explanation:

Project Status Area	Status (RYG)	Explanation
1. Overall Project Status		Core system (3 experts + arbitration + API + frontend integration) has been successfully completed and tested. System produces structured evaluation outputs.
2. Project Schedule		Project is on track for April 14 Capstone presentation. Only final polishing (UI + slides + documentation) remains.
3. Project Deliverables		Core deliverables completed, but model cards, evaluation metrics, and documentation need to be finalized for full submission quality.
4. Resources & Collaboration		All required resources (trained adapters, evaluation pipeline, API, frontend) are available and functioning. Team collaboration is effective.
5. Changes		Minor architecture adjustments were made (e.g., switching between local, ngrok, and sandbox deployment). These changes are controlled but require stabilization.
6. Communication		Regular updates and alignment maintained with team and project requirements. No major communication issues.

For status above, indicate **Red**, **Orange**, or **Green**:

- **Red**: Critical issues, serious risks to project, significant intervention must occur to achieve success, potential for stoppage of project activity. Project slipping by 5+ days, and resources uncommitted to meet deliverables
- **Orange**: Some major issues, moderate risk to project, must monitor closely, some internal or/and external dissatisfaction with progress. Project plan slipping by 2+ days.
- **Green**: No major issues, minimal risk to project, on target with expected outcomes, project on schedule, everyone satisfied with progress.

2. List All Completed Project Tasks:

- Trained three LoRA adapters (Governance, Threat, Behavioral)
- Built evaluation pipeline (`evaluate_system.py`) integrating all experts
- Implemented arbitration logic for final council decision
- Generated structured JSON outputs including risk levels and recommendations
- Developed FastAPI backend for model serving
- Connected frontend (Replit) to backend API
- Completed end-to-end testing of evaluation workflow
- Designed UI with expert selection and result display

3. List any concerns or issues that need the professor's involvement:

- Clarification on expected depth of Model Cards (how detailed they should be for grading)
- Confirmation whether quantitative evaluation metrics are required or qualitative evaluation is sufficient
- Guidance on presentation expectations (focus on system demo vs. technical depth)

4. Next series of tasks to complete:

- Complete Model Cards (fill all sections with meaningful content)
- Add evaluation results and performance metrics into documentation
- Improve frontend UI/UX (simplify layout, enhance visualization)
- Prepare final Capstone presentation slides
- Create demo script for presentation (10 min structured walkthrough)
- Ensure stable deployment (local + optional sandbox compatibility)

5. Sponsor Signoff

Sponsor indicates agreement with the above status report:

By (signature): Andres Fortino (signed) AFortino
Project Sponsor

Printed Name: Andrés Fortino
Please print in English

[Status Report C]

Project Status Report

Your Name: Yuze Wu

Project Title: UNICC

Date of report: 18th Of April

1. Project Status and Explanation:

Project Status Area	Status (RYG)	Explanation
1. Overall Project Status	Green	The integrated council-of-experts system has been completed, tested, and presented. The final product includes expert evaluation, deliberation, arbitration, API support, and a working front-end demo.
2. Project Schedule	Green	The project schedule has been successfully met. The Capstone presentation has been completed, and the project is now in final closeout and submission stage.
3. Project Deliverables	Green	Core deliverables have been completed, including the expert modules, arbitration logic, UI workflow, demo scenario, GitHub repository, and presentation materials.
4. Resources & Collaboration	Green	All required resources were available for final delivery. Team collaboration remained effective through integration, testing, and presentation.
5. Changes	Green	Minor final adjustments were made to improve UI flow, demo clarity, and deployment stability. These changes were completed without affecting timeline or scope.
6. Communication	Green	Communication with teammates and sponsor requirements remained consistent throughout the final phase. No major communication issues were encountered.

For status above, indicate **Red**, **Orange**, or **Green**:

- **Red**: Critical issues, serious risks to project, significant intervention must occur to achieve success, potential for stoppage of project activity. Project slipping by 5+ days, and resources uncommitted to meet deliverables
- **Orange**: Some major issues, moderate risk to project, must monitor closely, some internal or/and external dissatisfaction with progress. Project plan slipping by 2+ days.
- **Green**: No major issues, minimal risk to project, on target with expected outcomes, project on schedule, everyone satisfied with progress.

2. List All Completed Project Tasks:

- Completed integration of the three expert modules (Governance, Threat, Behavioral)
- Implemented council arbitration logic and final recommendation workflow
- Built and tested deliberation flow for expert disagreement handling
- Completed FastAPI backend and end-to-end API connectivity
- Finalized front-end interface for scenario input, expert review, deliberation, and council decision display
- Created and validated demo scenario for the final presentation
- Completed system testing across the full evaluation workflow
- Prepared GitHub repository, README, and supporting documentation for submission

Delivered final Capstone presentation with a working product demonstration

3. List any concerns or issues that need the professor's involvement:

- No major issues at this stage.
- Only final sponsor signature and administrative submission steps remain.

4. Next series of tasks to complete:

- Obtain sponsor signature on the final status report
- Submit the completed status report and final project materials
- Archive final documentation, presentation materials, and repository links for recordkeeping

5. Sponsor Signoff

Sponsor indicates agreement with the above status report:

By (signature): Andres Fortino
Project Sponsor

Printed Name: Andrés Fortino
Please print in English

Appendix I — Annotated Bibliography

1. National Institute of Standards and Technology. (2023). AI Risk Management Framework (AI RMF 1.0).

Abstract
ChatGPT provided summary:
This framework introduces a structured approach to managing AI risks through four core functions: map, measure, manage, and govern. It emphasizes continuous monitoring, lifecycle management, and organizational responsibility in AI deployment. The framework aims to help organizations identify and mitigate risks associated with AI systems.

Researcher comments
This framework is highly relevant because it provides the foundational structure for AI risk evaluation. However, it remains conceptual and does not offer concrete implementation methods. This limitation supports the need for a more operationalized system like the UNICC AI Safety platform.

2. European Commission. (2022). Proposal for a Regulation on Artificial Intelligence (AI Act).

Abstract
ChatGPT provided summary:
The EU AI Act introduces a risk-based classification system for AI systems, categorizing them into unacceptable, high-risk, and low-risk applications. It establishes regulatory requirements for high-risk AI systems, including transparency, documentation, and compliance standards. The framework aims to ensure safe and ethical AI deployment across the European Union.

Researcher comments
This regulation is important because it defines clear risk categories and governance expectations. However, it is designed for a specific legal jurisdiction and may not fully apply to global institutions like the UN. This highlights the need for adaptable and operational evaluation frameworks.

3. International Organization for Standardization. (2023). ISO/IEC 42001: Artificial Intelligence Management Systems.

Abstract
ChatGPT provided summary:
ISO/IEC 42001 establishes requirements for managing AI systems within organizations. It focuses on governance, accountability, documentation, and continuous improvement. The standard provides guidance on implementing AI management systems aligned with risk and compliance principles.

Researcher comments
This standard is useful for understanding organizational-level AI governance. However, it does not provide detailed technical methods for evaluating AI outputs. This gap reinforces the importance of integrating governance with executable testing systems.

4. Raji, I. D., et al. (2020). Closing the AI accountability gap. Proceedings of the ACM Conference on Fairness, Accountability, and Transparency.

Abstract
ChatGPT provided summary:
This paper discusses the lack of accountability in AI systems and proposes structured auditing processes across the AI lifecycle. It highlights the importance of documentation, transparency, and continuous monitoring. The study emphasizes that AI systems should be evaluated systematically rather than through ad hoc processes.

Researcher comments
This research directly supports the need for structured and repeatable evaluation workflows. It shows that current practices are inconsistent and lack transparency. This aligns with the motivation behind developing a unified AI safety evaluation system.

5. Bommasani, R., et al. (2021). On the opportunities and risks of foundation models. arXiv.

Abstract
ChatGPT provided summary:
This paper explores the capabilities and risks of foundation models, including large language models. It highlights issues such as bias, misuse, and lack of transparency. The study provides a comprehensive overview of how these models impact various industries.

Researcher comments
This paper is important because it explains why AI safety evaluation is necessary. It identifies multiple risks that need to be tested before deployment. These risks form the basis for designing structured evaluation systems.

6. Brundage, M., Avin, S., Wang, J., Belfield, H., Krueger, G., Hadfield, G., Khlaaf, H., Yang, J., Toner, H., Fong, R., Maharaj, T., Koh, P. W., Hooker, S., Leung, J., Trask, A., Bluemke, E., Lebensold, J., O'Keefe, C., Koren, M., … Anderljung, M. (2020). Toward trustworthy AI development: Mechanisms for supporting verifiable claims. arXiv.

Abstract
ChatGPT provided summary:
This paper examines how AI developers can make trustworthy and verifiable claims about the safety, fairness, privacy, and security of AI systems. It proposes a range of mechanisms, including audits, documentation, third-party review, and technical tools, to improve accountability in AI development. The study emphasizes that trust in AI should be supported by evidence rather than broad statements of intent.

Researcher comments
This paper is very useful for my research because it moves beyond general AI ethics principles and focuses on how organizations can demonstrate trustworthy development in practice. Its emphasis on verifiable claims directly supports the need for structured and auditable evaluation workflows. This is highly relevant to the UNICC project because our system also aims to make AI safety assessments more transparent, evidence-based, and operational.

7. Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines. Nature Machine Intelligence, 1(9), 389–399.

Abstract
ChatGPT provided summary:
This paper reviews a large number of global AI ethics guidelines and identifies common principles such as transparency, justice and fairness, non-maleficence, responsibility, and privacy. The authors show that although many organizations agree on broad ethical goals, there are important differences in interpretation and implementation. The study concludes that consensus on principles does not automatically lead to consistent governance practices.

Researcher comments
This source is important because it shows that AI governance already has a strong ethical

foundation at the principle level. However, it also makes clear that these principles are often too broad to guide actual system evaluation and deployment decisions. This supports my research by showing why an operational safety-testing framework is needed to translate high-level ethics into measurable and repeatable practice.

8. Mittelstadt, B. (2019). Principles alone cannot guarantee ethical AI. Nature Machine Intelligence, 1(11), 501–507.

Abstract
ChatGPT provided summary:
This article argues that high-level ethical principles such as fairness, accountability, and transparency are not sufficient by themselves to ensure ethical AI. The author explains that AI development lacks the professional norms, enforcement mechanisms, and institutional structures needed to translate principles into action. As a result, ethical AI requires practical tools, governance systems, and accountability mechanisms in addition to broad principles.

Researcher comments
This article is highly relevant because it clearly explains the gap between ethical intention and real implementation. It helps justify why my project cannot stop at discussing governance frameworks conceptually, but instead must focus on building a practical evaluation process. In the context of UNICC, this article supports the argument that AI safety requires executable mechanisms, not just policy language.

9. Morley, J., Machado, C. C. V., Burr, C., Cowls, J., Joshi, I., Taddeo, M., & Floridi, L. (2020). The ethics of AI in health care: A mapping review. Social Science & Medicine, 260, 113172.

Abstract
ChatGPT provided summary:
This paper reviews how AI ethics is being discussed and applied in the health care domain, with attention to themes such as fairness, transparency, accountability, privacy, and safety. It finds that ethical discussion is widespread, but practical implementation remains difficult and fragmented across institutions. The study highlights the challenge of turning broad ethical expectations into usable evaluation and oversight mechanisms.

Researcher comments
Although this paper is focused on health care, it is still valuable for my research because it illustrates a broader implementation problem that also appears in institutional AI settings. The article shows that organizations often understand what ethical AI should look like, but struggle with how to evaluate and govern it consistently. This directly supports my research goal of designing a more structured and usable AI safety evaluation workflow.

10. Morley, J., Floridi, L., Kinsey, L., & Elhalal, A. (2023). Operationalising AI ethics: Barriers, enablers and next steps. AI & Society, 38, 411–423.

Abstract

ChatGPT provided summary:

This paper examines why organizations struggle to put AI ethics into practice and identifies both barriers and enablers to operationalization. The authors discuss the difficulty of translating ethical principles into design practices, governance processes, and practical tools. They argue that moving from theory to implementation requires clearer methods, organizational support, and usable frameworks.

Researcher comments

This source is especially helpful because it focuses directly on the operationalization challenge, which is central to my project. It supports the idea that the main problem is not the absence of ethical principles, but the absence of actionable systems that allow organizations to apply them consistently. For my research, this article strengthens the argument for building an integrated AI safety evaluation platform rather than relying only on guidelines or manual review.

11. Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). "Why should I trust you?" Explaining the predictions of any classifier. Proceedings of the ACM SIGKDD Conference.

Abstract

ChatGPT provided summary:

This paper introduces LIME, a method for explaining the predictions of machine learning models. It provides local explanations that help users understand why a model makes a specific decision. The study emphasizes the importance of interpretability for building trust in AI systems.

Researcher comments

This paper is important because it highlights interpretability as a key requirement for trustworthy AI. In the context of my research, it supports the need for AI safety systems to provide not only evaluation results but also understandable explanations. This aligns with the UNICC project's goal of making safety decisions transparent and usable for non-technical stakeholders.

12. Doshi-Velez, F., & Kim, B. (2017). Towards a rigorous science of interpretable machine learning. arXiv.

Abstract
ChatGPT provided summary:
This paper explores the concept of interpretability in machine learning and proposes a framework for evaluating interpretability methods. It discusses different types of explanations and the importance of aligning them with user needs. The study highlights the challenges of measuring interpretability in a rigorous and consistent way.

Researcher comments
This research is useful because it shows that interpretability itself is complex and requires structured evaluation. It supports my project by emphasizing that AI evaluation systems must carefully design how explanations are generated and presented. This is particularly important for institutional settings like UNICC, where decisions must be clearly justified.

13. Weidinger, L., et al. (2021). Ethical and social risks of harm from language models. arXiv.

Abstract
ChatGPT provided summary:
This paper identifies and categorizes various risks associated with language models, including bias, misinformation, misuse, and harmful content generation. It provides a structured taxonomy of risks that can arise from deploying large-scale AI systems. The study emphasizes the importance of proactively identifying and mitigating these risks.

Researcher comments
This paper is highly relevant because it clearly defines the types of risks that need to be evaluated in AI systems. It helps inform what dimensions should be included in a safety evaluation framework. For my research, it provides a foundation for designing structured testing criteria in the UNICC system.

14. Gehman, S., et al. (2020). RealToxicityPrompts: Evaluating neural toxic degeneration in language models. Findings of EMNLP.

Abstract
ChatGPT provided summary:
This paper introduces the RealToxicityPrompts dataset, which is designed to evaluate the tendency of language models to generate toxic outputs. It provides a benchmark for measuring harmful content generation under different conditions. The study highlights the need for systematic testing of model behavior.

Researcher comments
This paper is useful because it demonstrates how specific risks, such as toxicity, can be evaluated using structured benchmarks. However, it focuses on only one dimension of AI safety. This supports my research argument that a comprehensive evaluation system must integrate multiple types of tests rather than relying on isolated benchmarks.

15. Vidgen, B., et al. (2020). Learning from the worst: Dynamically generated datasets for hate speech detection. Proceedings of ACL.

Abstract
ChatGPT provided summary:
This paper presents a method for generating datasets to detect hate speech and harmful language in AI systems. It focuses on improving the detection of abusive and toxic content through dynamic data generation. The study contributes to improving robustness in content moderation systems.

Researcher comments
This paper contributes to understanding how datasets can be used to test harmful content detection in AI models. However, it does not address how these evaluations fit into a broader governance framework. This reinforces the need for integrated systems like the UNICC platform that combine multiple evaluation approaches into a unified workflow.

16. Perez, E., et al. (2022). Red teaming language models. arXiv.

Abstract
ChatGPT provided summary:

This paper explores the use of red teaming to identify vulnerabilities in large language models. It involves generating adversarial prompts to test model behavior under challenging or malicious conditions. The study shows that red teaming is effective in uncovering harmful outputs and policy violations.

Researcher comments
This paper is highly relevant because it demonstrates one of the core methods used in AI safety evaluation. It shows how adversarial testing can reveal risks that standard evaluation methods may miss. However, it also highlights scalability challenges, which supports the need for automated and integrated evaluation systems like the UNICC platform.

17. Ganguli, D., et al. (2022). Red teaming language models with language models. arXiv.

Abstract
ChatGPT provided summary:
This paper introduces an automated approach to red teaming by using language models to generate adversarial test cases. It significantly improves the scalability of safety testing compared to manual methods. The study shows that AI-driven red teaming can systematically identify weaknesses in model behavior.

Researcher comments
This work is important because it addresses the scalability limitations of traditional red teaming. It supports the idea that AI systems can be used to evaluate other AI systems. However, it also raises concerns about evaluation bias and reliability, reinforcing the need for structured and controlled evaluation frameworks.

18. Zheng, L., et al. (2023). Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. arXiv.

Abstract
ChatGPT provided summary:
This paper evaluates the effectiveness of using large language models as judges to assess the quality of AI-generated responses. It introduces benchmarks such as MT-Bench and Chatbot

Arena to compare different models. The study finds that LLM-based evaluation is scalable but may suffer from biases and inconsistencies.

Researcher comments
This paper is directly relevant to my project because it validates the concept of using LLMs as evaluators. It supports the feasibility of scalable AI safety evaluation. However, the identified biases highlight the importance of designing structured evaluation criteria and combining multiple expert perspectives, which aligns with the UNICC Council approach.

19. Ji, Z., et al. (2023). Survey of hallucination in natural language generation.

ACM Computing Surveys.

Abstract
ChatGPT provided summary:
This paper provides a comprehensive survey of hallucination in natural language generation systems. It examines the causes, types, and impacts of incorrect or fabricated outputs. The study highlights the importance of detecting and mitigating hallucination in AI applications.

Researcher comments
This paper is important because hallucination is one of the key risks in AI systems, especially in high-stakes environments. It helps define a critical evaluation dimension for safety testing. For my research, it reinforces the need to include reliability and factual accuracy checks in the evaluation framework.

20. Meske, C., & Bunde, E. (2023). Transparency and explainability in AI systems:

The business perspective. Business & Information Systems Engineering.

Abstract
ChatGPT provided summary:
This paper examines the importance of transparency and explainability in AI systems from an organizational perspective. It highlights how explainable outputs improve user trust and decision-making. The study emphasizes the need for user-friendly interfaces that present AI results clearly.

Researcher comments
This source is valuable because it connects technical evaluation with user experience and decision-making. It supports the idea that AI safety systems must not only evaluate risks but also present results in a clear and interpretable way. This aligns closely with the interface design goals of the UNICC project.